

ASSIGNMENT 4 : Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.preprocessing import LabelEncoder

data = pd.read_csv("titanic.csv")

data['Age'] = data['Age'].fillna(data['Age'].mean())
data['Fare'] = data['Fare'].fillna(data['Fare'].mean())

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data['Age'] = pd.cut(data['Age'], bins=4, labels=False)

data['Fare'] = pd.cut(data['Fare'], bins=4, labels=False)

X = data[['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare']]
y = data['Survived']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

model = MultinomialNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Multinomial Naïve Bayes Result")

print("Accuracy:", accuracy)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)

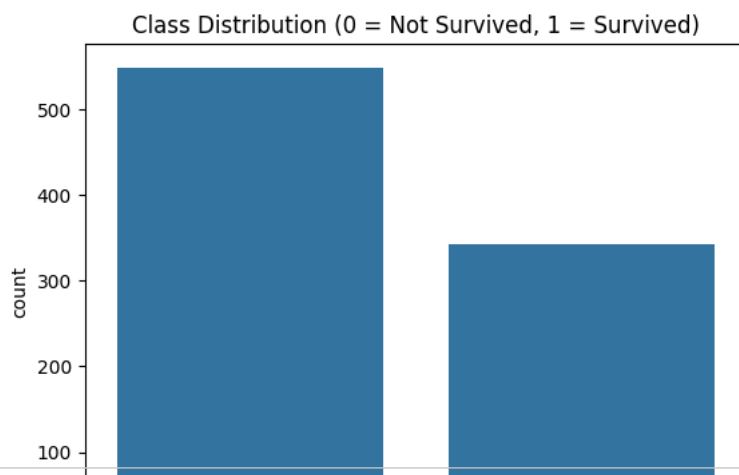
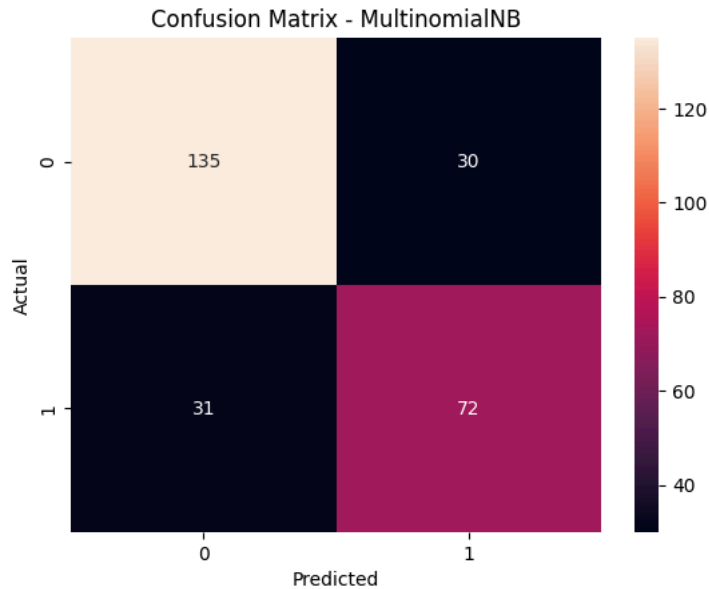
plt.figure()
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix - MultinomialNB")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

plt.figure()
sns.countplot(x='Survived', data=data)
plt.title("Class Distribution (0 = Not Survived, 1 = Survived)")
plt.show()
```

Multinomial Naïve Bayes Result
Accuracy: 0.7723880597014925

Classification Report:

	precision	recall	f1-score	support
0	0.81	0.82	0.82	165
1	0.71	0.70	0.70	103
accuracy			0.77	268
macro avg	0.76	0.76	0.76	268
weighted avg	0.77	0.77	0.77	268

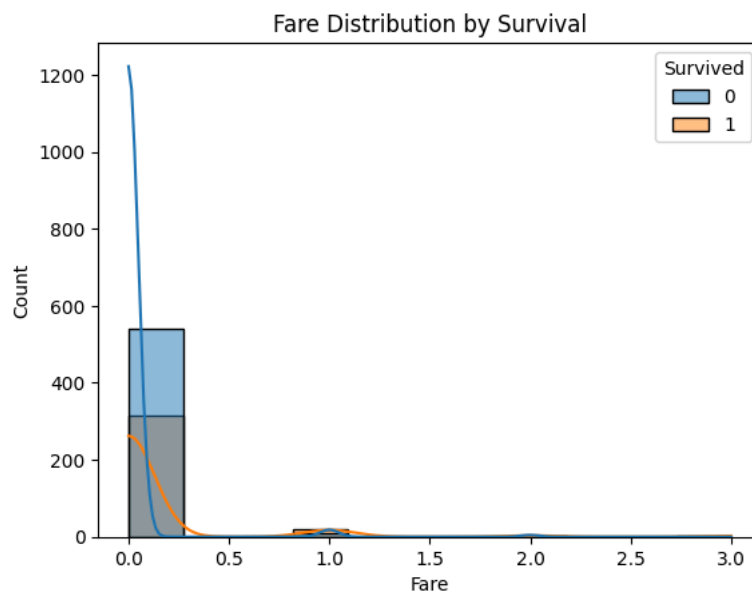
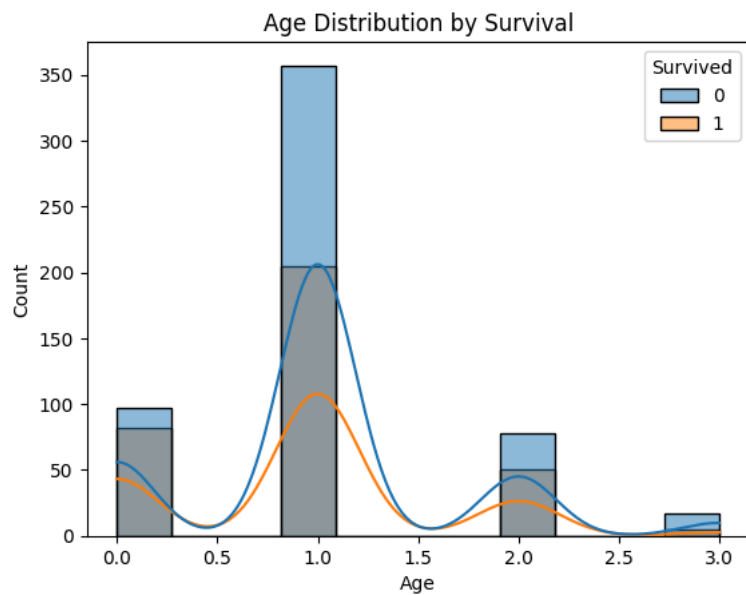


```
plt.figure()
sns.histplot(data=data, x='Age', hue='Survived', kde=True)
plt.title("Age Distribution by Survival")
plt.show()
```

```
plt.figure()
sns.histplot(data=data, x='Fare', hue='Survived', kde=True)
plt.title("Fare Distribution by Survival")
plt.show()
```

```
sample_data = np.array([[3, 0, 25, 0, 0, 7.25]])
prediction = model.predict(sample_data)
```

```
if prediction[0] == 1:
    print("\nManual Test Prediction: Survived")
else:
    print("\nManual Test Prediction: Not Survived")
```



Manual Test Prediction: Survived

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names
warnings.warn()